

Heat-flux affinity sweep and Gaussian crossover

Frozen $n = 10$ Cartesian experiment with equilibrium amendment, 2026-09-04

Scope and frozen estimator

The transported heat in a window is

$$Q = \frac{Q_L - Q_R}{2}, \quad \Delta\beta = \frac{1}{T_R} - \frac{1}{T_L}.$$

Non-overlapping windows are formed within each of 128 independent streams. The directly sampled fluctuation-relation slope is obtained from raw matched bins of $\log[p(Q)/p(-Q)]$ against Q . The fixed rule is $\Delta Q = \text{std}(Q)/20$, at least 10 raw observations on each side of every accepted bin pair, and the largest contiguous reliable block that straddles zero. No plus-four points or extrapolated densities enter the fit. Stream bootstrap intervals resample the 128 complete streams.

The Gaussian bulk diagnostics are

$$a_{\text{Gauss}} = \frac{2\langle Q \rangle}{\text{Var}(Q)}, \quad G_{\text{FT}} = \frac{\text{Var}(Q)}{2\langle Q \rangle / \Delta\beta}.$$

They characterize the CLT crossover but are not substitutes for a directly sampled negative tail.

Provenance

Experiment repository commit: `e170205c5d9a3d9b2bfca3714a4447fa965a1e40`.

Production source commit: `1905cf4e606a4a7f4dd8930caa64bd4cc861e9d4`.

Frozen source SHA-256: `98e7f8f5f915c8ce02bd8aa10722025c09fd739184b981961692869c9356c0d3`.

Driven binary SHA-256: `4c4880d721733897d200f2601690da873f5b226aaa1013df23df2351c1dfe7d1`.

Equilibrium binary SHA-256: `5fdcd596f0b4da39357a7217fb8b475f14e7b381da08a1208351fccbc008080e`.

The four driven runs and the equilibrium amendment use $n = 10$, $\gamma = 0.1$, $dt = 5 \times 10^{-4}$, burn-in 500, base block duration 20, 8 SIMD batches, 16 lanes, 7,813 blocks per stream, and 1,000,064 blocks per case.

Exact production commands.

```
dbeta_0p027972 /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪ sample 6.5 5.5 10 8 500 20 7813 0.0005 2026090401 2
↪ /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p027972_n10 5
dbeta_0p057143 /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪ sample 7 5 10 8 500 20 7813 0.0005 2026090402 2
↪ /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p057143_n10 5
dbeta_0p125000 /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪ sample 8 4 10 8 500 20 7813 0.0005 2026090403 2
↪ /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p125000_n10 5
dbeta_0p222222 /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪ sample 9 3 10 8 500 20 7813 0.0005 2026090404 2
↪ /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p222222_n10 5
dbeta_0p000000
↪ /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity_equilibrium
↪ sample 6 6 10 8 500 20 7813 0.0005 2026090405 2
↪ /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p000000_n10 5
```

Input hashes.

Case	rows	SHA-256
dbeta_0p027972	1000064	8b44be37ea8f566fb3862b6cd796dcfcf433be0888752bab876754c84d4c421
dbeta_0p057143	1000064	382496fcd194154b0f6d486f2d4da9de80d2d0791f94cb0f3e695d8e666a29cc
dbeta_0p125000	1000064	0404492717ce335c1e2eff7970d5d514435fabec9b91604fa92b7d4846cdf685
dbeta_0p222222	1000064	342f3783b40db98263cd685c8ab4e531c9d83842db6762b0bd0a06061642958d
dbeta_0p400000	1000064	a23806e82f5514a9c3375d10a6644946b6b57d0efe8452b3bbf397b6230f9929
dbeta_0p000000	1000064	7fd6827f12b42844de1142613af124702ca5535ba63f3c54ab71006b256aef4d

Full absolute input paths are retained in `analysis/input_hashes.csv`.

Outcome under the frozen gates

The full-sample $1/t$ intercepts give $a_\infty/\Delta\beta = 1.0175, 1.0153, 1.0220$ at $\Delta\beta = 0.027972, 0.057143, 0.125$, respectively. These point estimates lie close to the FT reference, but none of the cases reaches the predeclared threshold of 800 valid joint stream-bootstrap intercepts. Consequently every nonzero-affinity FT verdict is **UNRESOLVED**; the point estimates alone are not promoted to a pass. For the equilibrium control, the full-sample intercept is 7.80×10^{-5} . Its $t = 20$ direct slope is 1.63×10^{-4} with stream-bootstrap 95% interval $[-8.98 \times 10^{-5}, 4.18 \times 10^{-4}]$, which contains zero. The joint long-time interval again fails the 800-replicate gate, so the formal equilibrium extrapolation remains **UNRESOLVED**. The Gaussian bulk approaches the FT value only at weak drive: $a_{\text{Gauss}}/\Delta\beta = 0.960, 0.931, 0.745, 0.515, 0.301$ as the affinity increases. At $t = 160$, the negative counts fall from 20,136 and 2,997 in the two weakest driven cases to 2, 0, and 0 in the three stronger cases. This directly records the loss of two-tail support rather than replacing it by a Gaussian estimate.

Raw window statistics and two-tail fits

dbeta_0p027972: $(T_L, T_R) = (6.5, 5.5)$, $\Delta\beta = 0.027972$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	8.9368	25.8599	0.1348	0.3798	366513	0.02720	$[-, -]$	0.9978	0.02673 / 1.0466
40	499968	17.8742	36.3277	0.1019	0.2084	155995	0.02758	$[-, -]$	0.9979	0.02709 / 1.0326
80	249984	35.7483	51.2908	0.0770	0.0995	60756	0.02781	$[0.02736, 0.02818]$	0.9979	0.02718 / 1.0292
160	124928	71.4974	72.6266	0.0512	0.0388	20136	0.02706	$[0.02645, 0.02754]$	0.9951	0.02711 / 1.0318
320	62464	142.9947	102.6171	0.0433	0.0109	5109	0.02778	$[-, -]$	0.9944	0.02716 / 1.0299
640	31232	285.9894	145.9539	0.0505	0.0103	764	0.03036	$[-, -]$	0.9704	0.02685 / 1.0418

dbeta_0p057143: $(T_L, T_R) = (7.0, 5.0)$, $\Delta\beta = 0.057143$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	17.7373	26.2279	0.2518	0.4064	248420	0.05487	$[0.05446, 0.05523]$	0.9993	0.05157 / 1.1081
40	499968	35.4752	36.8119	0.1869	0.2167	82050	0.05602	$[-, -]$	0.9991	0.05236 / 1.0914
80	249984	70.9503	51.9375	0.1401	0.0936	20447	0.05651	$[-, -]$	0.9986	0.05260 / 1.0863
160	124928	141.8966	73.3519	0.0951	0.0409	2997	0.05614	$[-, -]$	0.9928	0.05274 / 1.0834
320	62464	283.7931	103.2913	0.0737	0.0348	150	0.05978	$[-, -]$	0.6020	0.05320 / 1.0741
640	31232	567.5862	145.5179	0.0377	0.0130	1	UNRES.	—	—	0.05361 / 1.0659

dbeta_0p125000: $(T_L, T_R) = (8.0, 4.0)$, $\Delta\beta = 0.125000$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	34.8992	27.7171	0.4529	0.5303	88761	0.11337	$[-, -]$	0.9995	0.09086 / 1.3758
40	499968	69.7973	38.8933	0.3447	0.2789	12744	0.11966	$[-, -]$	0.9975	0.09228 / 1.3545
80	249984	139.5947	54.7524	0.2520	0.1553	661	0.12460	$[-, -]$	0.9866	0.09313 / 1.3422
160	124928	279.1903	77.2501	0.1806	0.0808	2	UNRES.	—	—	0.09357 / 1.3359
320	62464	558.3805	109.0295	0.1323	0.0777	0	UNRES.	—	—	0.09394 / 1.3306
640	31232	1116.7611	153.4323	0.1020	0.0404	0	UNRES.	—	—	0.09488 / 1.3175

dbeta_0p222222: $(T_L, T_R) = (9.0, 3.0)$, $\Delta\beta = 0.222222$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	50.8282	29.9660	0.5614	0.5940	24798	0.17618	$[0.17407, 0.17782]$	0.9994	0.11321 / 1.9630
40	499968	101.6576	42.1593	0.4311	0.3267	985	0.19148	$[-, -]$	0.9901	0.11439 / 1.9427
80	249984	203.3151	59.3516	0.3233	0.1607	3	UNRES.	—	—	0.11543 / 1.9251
160	124928	406.6254	83.9248	0.2323	0.0889	0	UNRES.	—	—	0.11546 / 1.9246
320	62464	813.2508	119.0787	0.1714	0.0688	0	UNRES.	—	—	0.11471 / 1.9373
640	31232	1626.5015	168.5450	0.0984	0.0109	0	UNRES.	—	—	0.11451 / 1.9406

dbeta_0p400000: $(T_L, T_R) = (10.0, 2.0)$, $\Delta\beta = 0.400000$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	64.2424	32.6857	0.6006	0.6213	7941	0.21573	[-, -]	0.9985	0.12026 / 3.3260
40	499968	128.4849	45.9342	0.4603	0.3523	80	UNRES.	-	-	0.12179 / 3.2844
80	249984	256.9699	64.7162	0.3377	0.1853	0	UNRES.	-	-	0.12271 / 3.2597
160	124928	513.9476	91.2279	0.2361	0.0836	0	UNRES.	-	-	0.12351 / 3.2387
320	62464	1027.8951	128.6532	0.1634	0.0240	0	UNRES.	-	-	0.12420 / 3.2205
640	31232	2055.7903	182.2058	0.1061	0.0165	0	UNRES.	-	-	0.12385 / 3.2298

dbeta_0p000000: $(T_L, T_R) = (6.0, 6.0)$, $\Delta\beta = 0$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	0.0306	25.7283	-0.0027	0.3783	499786	1.63012e-04	[-8.97958e-05, 4.18259e-04]	0.0214	9.24910e-05 / -
40	499968	0.0610	36.1107	-0.0068	0.2064	249282	-1.10678e-05	[-, -]	1.0247e-04	9.34879e-05 / -
80	249984	0.1219	50.8838	-0.0029	0.1190	124666	3.89544e-05	[-, -]	0.0010	9.41668e-05 / -
160	124928	0.2379	71.8208	-3.2658e-04	0.0612	62425	2.26664e-04	[-, -]	0.0551	9.22287e-05 / -
320	62464	0.4757	101.3193	-0.0103	0.0283	31032	-2.82407e-06	[-, -]	6.7076e-06	9.26856e-05 / -
640	31232	0.9515	143.2340	0.0089	-0.0130	15544	1.20689e-04	[-, -]	0.0124	9.27544e-05 / -

Long-time extrapolation

Case	$\Delta\beta$	resolved t	a_∞	95% CI	$a_\infty/\Delta\beta$	ratio CI	status
dbeta_0p027972	0.027972	20;40;80;160;320;640	0.02846	[-, -]	-	[-, -]	UNRESOLVED
dbeta_0p057143	0.057143	20;40;80;160;320	0.05802	[-, -]	-	[-, -]	UNRESOLVED
dbeta_0p125000	0.125000	20;40;80	0.12775	[-, -]	-	[-, -]	UNRESOLVED
dbeta_0p222222	0.222222	20;40	-	[-, -]	-	[-, -]	UNRESOLVED
dbeta_0p400000	0.400000	20	-	[-, -]	-	[-, -]	UNRESOLVED
dbeta_0p000000	0	20;40;80;160;320;640	7.80343e-05	[-, -]	-	[-, -]	UNRESOLVED

Crossover summary

$\Delta\beta$	$a_\infty/\Delta\beta$	ratio CI	$a_G/\Delta\beta$	$G_{FT}(640)$	skew(160)	ex.kurt(160)	$n_-(160)$	status
0	-	[-, -]	-	-	-3.2658e-04	0.0612	62425	UNRESOLVED
0.027972	-	[-, -]	0.9599	1.0418	0.0512	0.0388	20136	UNRESOLVED
0.057143	-	[-, -]	0.9310	1.0659	0.0951	0.0409	2997	UNRESOLVED
0.125000	-	[-, -]	0.7450	1.3175	0.1806	0.0808	2	UNRESOLVED
0.222222	-	[-, -]	0.5147	1.9406	0.2323	0.0889	0	UNRESOLVED
0.400000	-	[-, -]	0.3007	3.2298	0.2361	0.0836	0	UNRESOLVED

Figures

Equilibrium row and claim boundary

The added (6, 6) production uses the same $n = 10$ dynamics, integration step, burn-in, stream count, and block count as the driven cases. At $\Delta\beta = 0$, ratios that divide by $\Delta\beta$ and G_{FT} are undefined and are reported as dashes. The direct asymmetry-slope reference is zero; its extrapolated confidence interval is tested against zero rather than against a ratio of one.

The complete raw positive/negative counts used by every fit are in `analysis/symmetric_bin_raw_counts.csv`; all per-window statistics are in `analysis/window_summary.csv`. A case labelled `CONSISTENT_WITH_FT` means only that the predeclared extrapolated confidence interval includes the reference. It is not a proof. A case whose interval excludes the reference is labelled `FAIL`; insufficient raw two-tail support remains `UNRESOLVED`. The label `CONSISTENT_WITH_EQUILIBRIUM` has the corresponding limited meaning for the zero-affinity control.

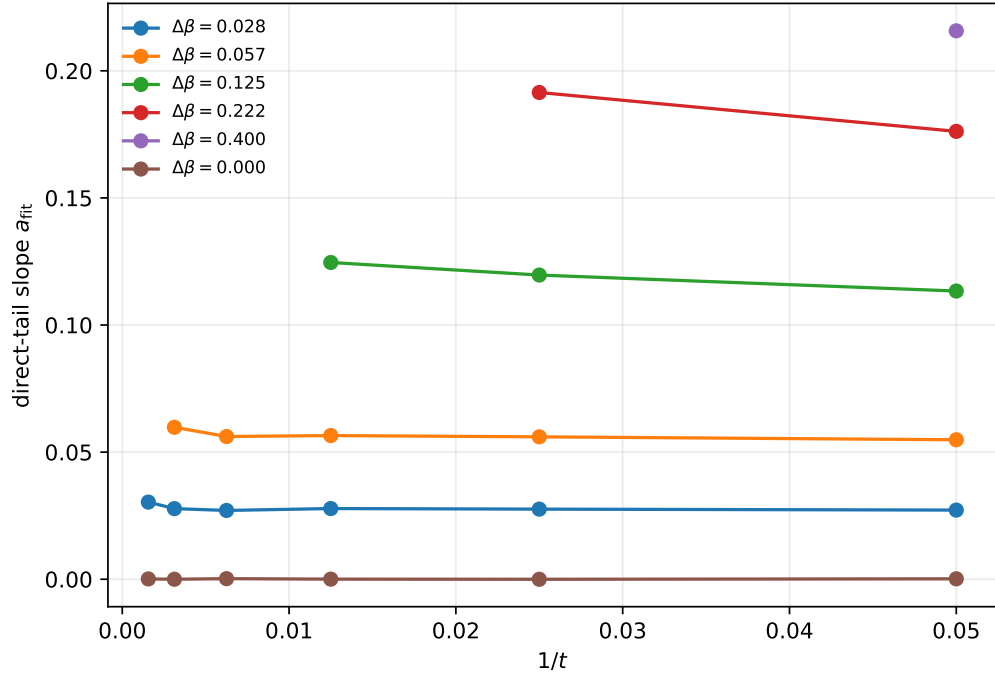


Figure 1: Direct matched-bin slopes against inverse window duration.

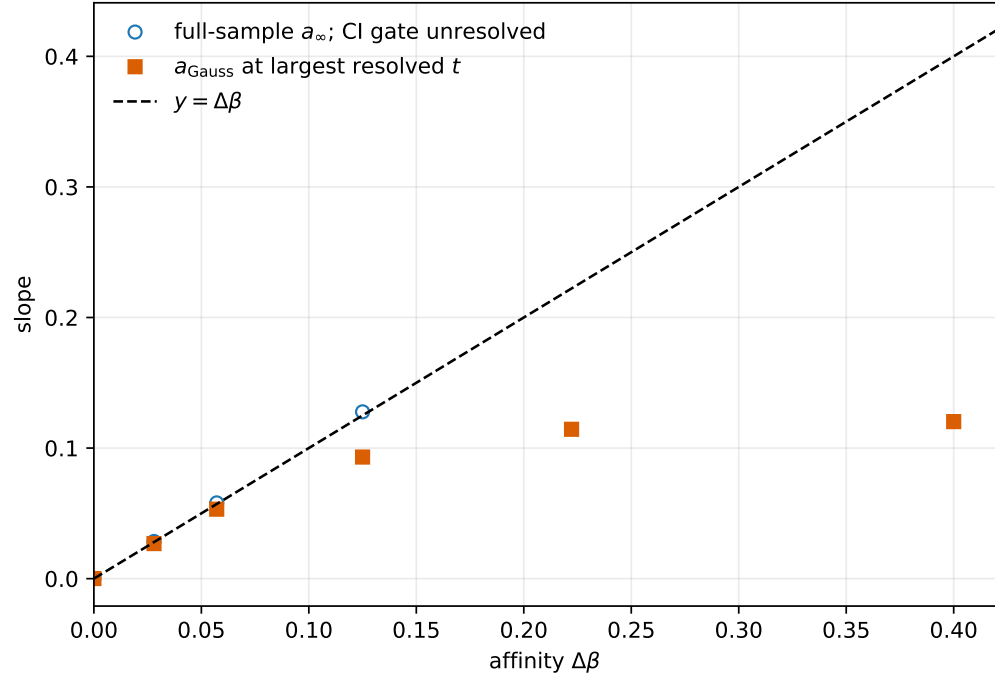


Figure 2: Extrapolated direct-tail slopes and Gaussian bulk slopes against affinity.

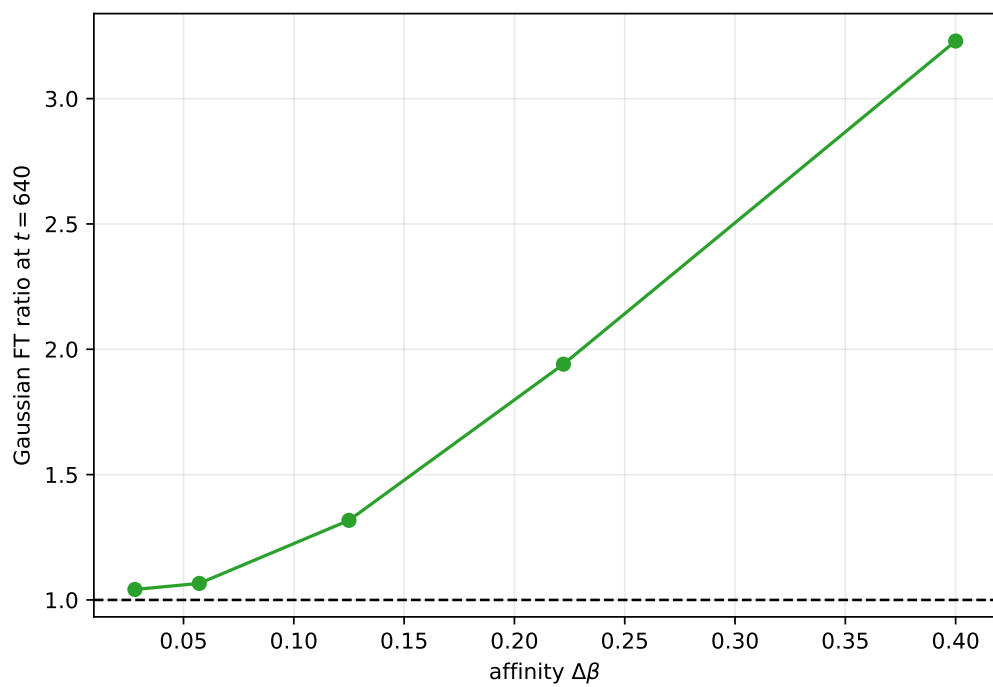


Figure 3: Gaussian FT ratio at $t = 640$ across affinity.